

Fixed Point Representation

1) 12.75 with 4 integer bits and 4 fractional bits:

Integer part: 12

12 in binary $\rightarrow$ **1100**

Fractional part: 0.75

Multiply by 2 repeatedly:

- $0.75 \times 2 = 1.5 \rightarrow$ **1**
- $0.5 \times 2 = 1.0 \rightarrow$ **1**
- $0.0 \times 2 = 0 \rightarrow$ **0**
- $0.0 \times 2 = 0 \rightarrow$ **0**

Fractional bits $\rightarrow$ **1100**

Answer=1100.1100

2) 5.125 with 3 integer bits and 5 fractional bits

Integer part: 5

5 in binary $\rightarrow$ **101**

Fractional part: 0.125

- $0.125 \times 2 = 0.25 \rightarrow$ **0**
- $0.25 \times 2 = 0.5 \rightarrow$ **0**
- $0.5 \times 2 = 1.0 \rightarrow$ **1**
- $0.0 \times 2 = 0 \rightarrow$ **0**
- $0.0 \times 2 = 0 \rightarrow$ **0**

Fractional bits $\rightarrow$ **00100**

Answer=101.00100

3) 7.5 with 4 integer bits and 4 fractional bits

Integer part: 7

7 in binary $\rightarrow$ **0111**

Fractional part: 0.5

- $0.5 \times 2 = 1.0 \rightarrow$ **1**
- $0.0 \times 2 = 0 \rightarrow$ **0**
- $0.0 \times 2 = 0 \rightarrow$ **0**
- $0.0 \times 2 = 0 \rightarrow$ **0**

Fractional bits $\rightarrow$ **1000**

Answer=0111.1000